6.5.1 Pair of Lines Passing through the Origin

We first consider a simple case. Both the lines in this pair pass through the origin. Thus, their equations can be written as

$$y - m_1 x = 0$$
$$y - m_2 x = 0$$

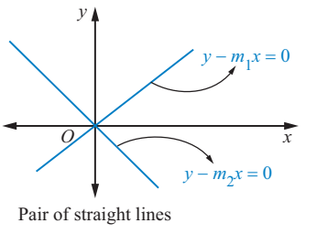


Figure 6.49

Combined equation of these two lines is

$$\begin{aligned} (y - m_1 x)(y - m_2 x) &= 0 \qquad (6.29) \\ y^2 - (m_1 + m_2) xy + m_1 m_2 x^2 &= 0 \end{aligned}$$

The above equation suggests that the general equation of a pair of straight lines passing through the origin with slopes m_1 and m_2, $ax^2 + 2hxy + by^2 = 0$ is **a homogenous equation of degree two**, implying that the degree of each term is 2.

Nature of the homogenous equations tells us whether the lines pass through the origin.

Example 6.33 Separate the equations $5x^2 + 6xy + y^2 = 0$

Solution:

We factorize this equation straight away as

$$\begin{aligned} 5x^2 + 6xy + y^2 &= 0 \\ 5x^2 + 5xy + xy + y^2 &= 0 \\ 5x(x + y) + y(x + y) &= 0 \\ (5x + y)(x + y) &= 0 \end{aligned}$$

So that the lines are $5x + y = 0$, and $x + y = 0$

Alternate method: since the given equation is a homogeneous equation, divide the given equation

$$\begin{aligned} 5x^2 + 6xy + y^2 &= 0 \text{ by } x^2 \\ \text{We get } 5 + 6\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2 &= 0 \\ \text{Substitute } \frac{y}{x} &= m \text{ (slope of the lines for homogenous equation)} \end{aligned}$$

The above equation becomes $m^2 + 6m + 5 = 0$

$$\begin{aligned} \text{Factorizing, we get} \qquad (m + 1)(m + 5) &= 0 \\ \Rightarrow m = -1, \quad m &= -5 \\ \Rightarrow \frac{y}{x} = -1, \quad \frac{y}{x} &= -5 \end{aligned}$$

That is, the lines are $x + y = 0, \quad 5x + y = 0$

Example 6.34 If exists, find the straight lines by separating the equations $2x^2 + 2xy + y^2 = 0$.

Solution:

Since the given equation is a homogeneous equation, divide the given equation $2x^2 + 2xy + y^2 = 0$ by x^2 and substituting $\frac{y}{x} = m$